FEASIBILITY MEMO

To:	Dr. De Freitas & Emergency Department Board, Mercer General Hospital
From:	Eliana Dookhoo
Date:	4 July 2026
Re:	Feasibility Assessment — Yale EMMLC ED Triage Dataset (AI-Assisted Triage Pilot)

1. One-Sentence Verdict

This dataset is feasible to build a triage prediction model from; it is large, complete, and clinically rich, but deployment at Mercer General requires careful caveats around population generalisability, class imbalance at the extremes of acuity, and a small number of physiologically implausible outlier values that must be resolved before modelling begins.

2. Dataset Executive Summary

Parameter	Value
Cohort Scale	55,121 unique ED presentation records
Feature Dimensionality	225 clinical variables — 7 vital signs, 200 chief complaint binary flags, 18 demographic/operational columns
Target Variable	ESI Triage Level (1–5); 1 = most critical — zero missing labels
ESI Class Profile	ESI 3 = 49.0% (n=27,010) | ESI 2 = 32.5% (n=17,924) | ESI 4 = 16.1% (n=8,896) | ESI 5 = 2.2% (n=1,214) | ESI 1 = 0.1% (n=77)
Disposition Split	Discharged: 62.7% (n=34,565) | Admitted: 37.3% (n=20,556)
Age Range	18–107 years (mean 55.3, SD 19.5) — adults only
Gender	Female: 57.6% | Male: 42.4%
Race	White/Caucasian: 53.4% | Black/African American: 29.0% | Other: 16.4% | Refused/Unknown: 1.2%
Missingness	0% across all structured columns — dataset was pre-cleaned before release
Primary Setting	Single academic medical centre — Yale New Haven Hospital, USA (not Caribbean)

3. Data Quality Dashboard

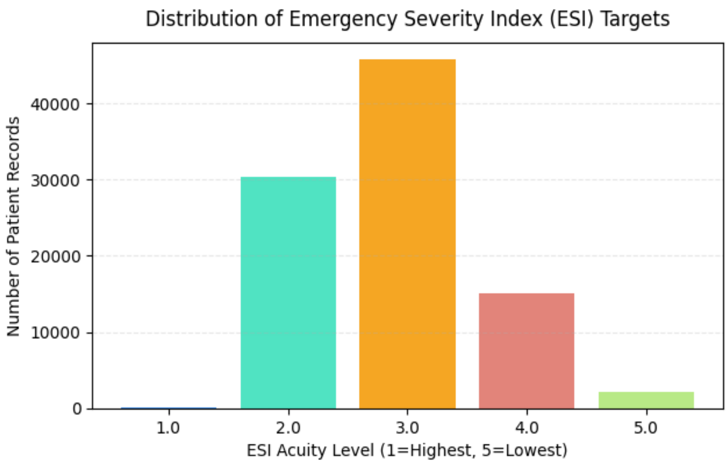

Figure 1: *Distribution of ESI Triage Levels. ESI 3 dominates at 49%; ESI 1 (most critical) accounts for only 77 patients (0.1%).*

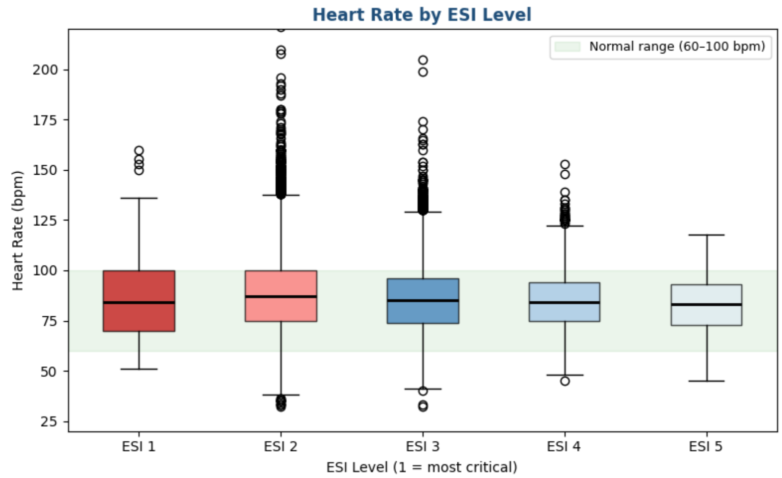

Figure 2: *Heart Rate by ESI Level. ESI 1 patients show a higher and wider IQR, consistent with haemodynamic instability. ESI 2–5 medians fall within the normal 60–100 bpm band (green)*

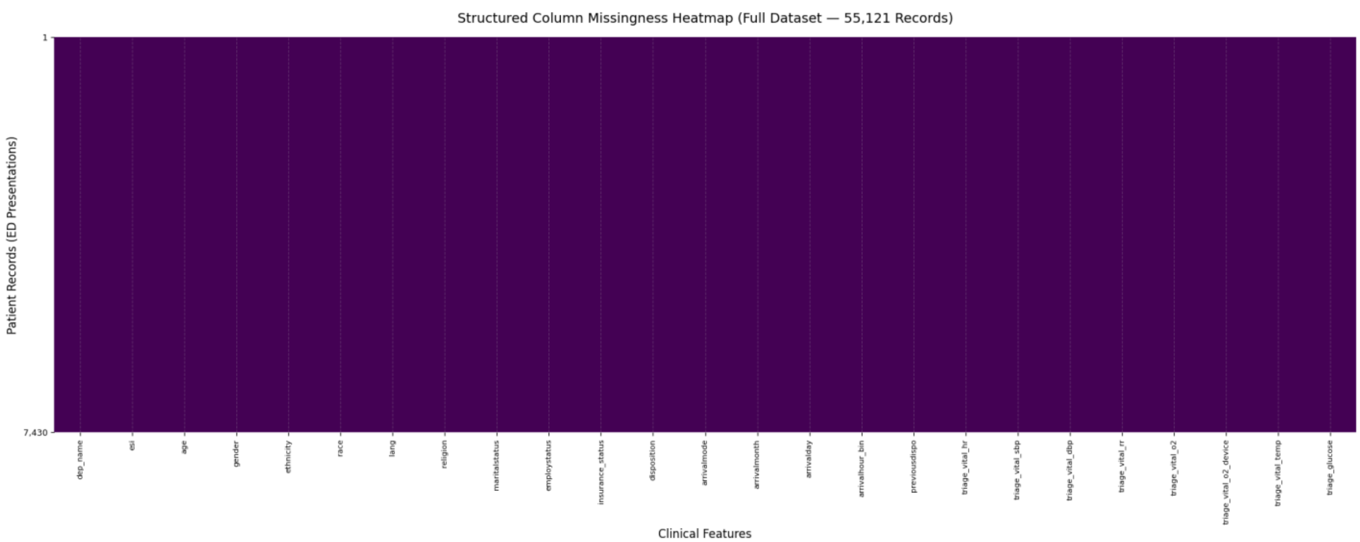

Figure 3: *Front-Door Vitals Missingness Heatmap. All cells dark (present) - zero missingness in vital signs. This confirms the dataset was pre-cleaned; Mercer's own EHR data will have gaps.*

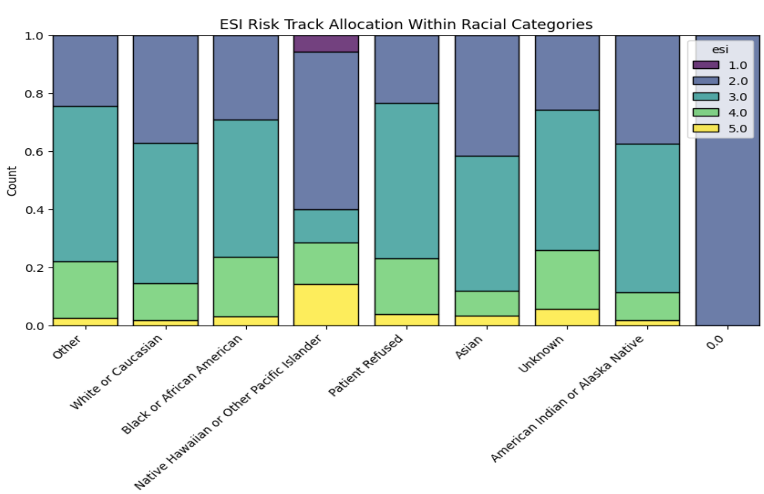

Figure 4: *ESI Risk Track Allocation Within Racial Categories. Black/African American patients are allocated to lower-urgency ESI 4 at 20.5% versus 12.8% for White/Caucasian patients - a 7.7 percentage point disparity warranting a formal fairness audit before deployment.*

4. Top 3 Data Quality Concerns

Quality Concern 1 – Severe Chief Complaint Sparsity

149 out of 200 diagnostic complaint flags (74.5%) have a prevalence below 0.5%, meaning they fire in fewer than 1 in 200 patient encounters. High-dimensional near-zero-variance features add noise without signal and risk inflating apparent model complexity. Mitigation: complaint flags below 0.5% prevalence will be dropped before modelling; the remaining 51 usable flags still capture all clinically significant presentations.

Quality Concern 2 – Vital Sign Estimation Noise (Respiratory Rate)

Respiratory rate shows an identical median of exactly 18 breaths per minute across all five ESI acuity classes, a pattern inconsistent with genuine physiological variance. This strongly suggests a triage data-entry shortcut where nurses defaulted to 18 bpm rather than measuring. As respiratory rate is clinically the earliest indicator of patient deterioration, this artefact could mask the feature's true predictive value. Mitigation: flag RR as unreliable for modelling; consider weighting it lower or excluding it from the initial baseline model in Week 6.

Quality Concern 3 – Acuity-Linked Data Completeness Variance

While all structured columns show zero overall missingness in this pre-cleaned release, glucose measurements in real EHR data track selectively by acuity; lower-acuity patients are less likely to have fingerstick glucose recorded at triage. This non-random missingness pattern means any imputation strategy must be informed by clinical context rather than applied uniformly. Mitigation: when applying this pipeline to Mercer General's live data, missingness flags will be retained as explicit model features per Naemi et al. (2021).

5. Top 3 Reasons to Proceed

Reason 1 – Strong Predictor Separation Signals

Oxygen saturation (triage_vital_o2) demonstrates the strongest linear correlation with ESI level (r = +0.178): low O2 reliably tracks with high acuity. Heart rate (Figure 3) shows visually clear separation at ESI 1, with a wider IQR and higher median consistent with haemodynamic instability. These physiologically coherent signal patterns confirm that the data contains genuine clinical information rather than noise.

Reason 2 – Sufficient Sample Depth for Safe Modelling

With 55,121 observations across five acuity classes, the dataset provides sufficient statistical power to train ensemble models (Random Forest, XGBoost) and evaluate performance across subgroups without risk of validation collapse. Even the smallest usable class, ESI 5 (n=1,214), exceeds the minimum sample size required for robust k-fold cross-validation. The zero-missingness structure further reduces preprocessing uncertainty in the baseline model.

Reason 3 – High Clinical Coherence in Key Complaint Flags

Chief complaint flags capture clinically meaningful signal where it matters most: 59.1% of patients presenting with chest pain (cc_chestpain) were correctly streamed into ESI 2, the emergent track. This level of complaint-to-acuity coherence confirms that the binary flag design reflects real clinical triage logic, providing a solid foundation for NLP-enhanced modelling when free-text chief complaint notes become available from Mercer General's own EHR.

6. Algorithmic Guardrails & Caveats

- **Demographic disparity:** Figure 4 shows that Black/African American patients are allocated to ESI 4 at 20.5% versus 12.8% for White/Caucasian patients. Models will incorporate fairness constraints and a formal subgroup equity audit will be conducted before any deployment recommendation is made to the Ministry of Health.
- **Class imbalance:** ESI 3 represents 49.0% of all visits; ESI 1 only 77 patients (0.1%). A naive accuracy-optimising model will default to predicting ESI 3 for every patient. Mitigation: SMOTE resampling and class-weighted loss functions in Week 6; evaluation using F1 score and AUROC rather than raw accuracy.
- **Imputation limits:** Missing vital signs in Mercer's own data must be handled using robust column-wise medians bounded strictly by physiological limits (per WHO ETAT guidelines); not standard algorithmic interpolation, which could produce clinically impossible imputed values.
- **Population transfer gap:** This is a US academic hospital dataset (Yale New Haven, Connecticut; 53.4% White/Caucasian). External validation at Mercer General is required before clinical use. Yang et al. (2024) confirmed that direct deployment without local adaptation produces poor and highly variable performance.
- **Adults only:** Dataset excludes patients under 18. Mercer deployment must be scoped to adult presentations only until a paediatric dataset is available.
- **Temperature unit:** All temperature values are in Fahrenheit (normal: 97.0–99.5°F). Any downstream clinical interface must document this explicitly to avoid misinterpretation as Celsius.